

VIR-832M - Clamp-On type Ultrasonic Flow & Energy Meter

Smart. Built for Precision: The VIR-832M is a high-performance noninvasive ultrasonic transit-time flow and energy meter engineered for accurate measurement of liquid flow and thermal energy in HVAC systems. Whether you need cooling energy monitoring, hydronic system optimization, or reliable flow measurement in large pipes, the VIR-832M delivers precision without disruption



With Clamp On detectors:

- Clamp on mounting
- Ideal for large-diameter and critical infrastructure pipelines
- Accurate flow and thermal energy measurement for HVAC optimization
- Reliable performance in demanding industrial and commercial environments

Key Advantages

- Wide bi-directional flow measurement range

Seamless System Integration

- Modbus RTU
- Modbus TCP/IP
- BACnet/IP

- Measures both flow and heating/cooling energy with add on temp Sensors
- Wide bi-directional flow measurement range
- Measures both flow and heating/cooling energy with add on temp Sensors
- Clamp on temp sensors
- Backlit LCD display
- Advanced temperature compensation for higher accuracy
- Designed for harsh mechanical room

Flow Measurement

- Chilled and hot water systems
- Cooling towers and condenser loops
- Water-glycol mixtures
- Process water and compatible industrial liquids
- District cooling and heating

Product Data Sheet

VIR 832-M: Clamp On type Ultrasonic Flow & Heat Meter

	Accuracy	$\pm 1\%$, with options for $\pm 0.5\%$ $\pm 0.2\%$ ± 0.025 ft/s (0.008 m/s)
	Repeatability	Better than 0.2%
	Principle	Transit-time measuring principle
	Measurement Period	500ms
Main Unit	Display	LCD with backlight, display accumulated flow/heat, instantaneous flow/heat, velocity, time etc. Analog output: 4-20mA or 0-20mA current output. Impedance 0.1kw. Accuracy 0.1%
	Output	OCT output: Frequency signal (1-9999HZ) Relay output: Programmable (no signal, reverse flow etc.) RS 485 serial port
	Input	Three analog input
	RTD For Heat Meter only	Two 2-wire, 3-wire Pt100/Pt1000/Pt 500 RTD 12-bit inputs; Range of $-40\ldots 200^{\circ}\text{C}$; Clamp-on resistor kits available
	Other functions	Automatically record the totaliser data up to 5 years and 16 years Option
	Energy total (Heat Meters)	British Thermal Unit (Btu), KWH
	Heat/cooling rate (Heat Meters)	Btu/hour, Kilowatts
	Power loss mode	The power-on time and corresponding flow rate of the last 64 power on and off events. Allow manual or automatic flow loss compensation
	Material	Steel, Stainless steel, Cast iron, Cement pipe, Copper, PVC, Aluminium, FRP etc. Liner is allowed
	Size	15-6000mm
Pipe	Straight pipe section	In the upstream it must be beyond 10D, in the downstream it must be beyond 5D. In the upstream the length must be beyond 30D from the access of the pump. (D Stands for pipe diameter)
	Types	Water, sea water, industrial sewage, acid and alkali liquid, alcohol, beer, all kinds of oils which can transmit ultrasonic single uniform liquid
Liquid	Temperature	Standard : -10°C - 160°C
	Turbidity	Less than 10000ppm, with a little bubble
	Flow Direction	Bi-directional measuring, net flow/heat measuring
	Ambient Temperature	Main Unit: $-4\ldots 140^{\circ}\text{F}$ ($-20\ldots 60^{\circ}\text{C}$)
Environment	Altitude Restriction	Up to 2000 m (6561 ft)
	Humidity	Main Units: 0...85%, non-condensing Transducer : water-immersible, water depth less than 3m
Cable	Twisted pair line, standard length of 20m, can be extended to 500m (not recommended); Contact the manufacturer for longer cable requirement. RS-485 interface, transmission distance up to 1000m	
Power Supply	DC24V	
Power	Less than 1.5W	
Protocols	MODBUS-RTU Protocol Standard. Option of BACnet -IP .	

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